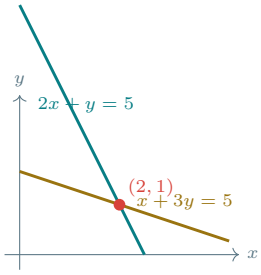
LINEAR ALGEBRA LADDER 1 OF 9

# Vectors, combinations and the two pictures

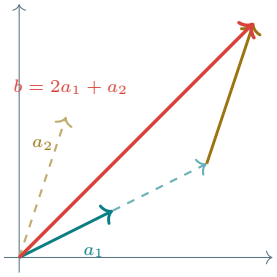
A system of equations is both a pair of lines and a combination of columns.

## 1 / THE RULE AND THE PICTURE

### The system $2x + y = 5$, $x + 3y = 5$ is two lines meeting, and also the vector $b$ built from two columns




row picture




column picture

Reading the equations by rows gives two lines through $(2, 1)$. Reading them by columns asks how much of each column is needed to reach $b$:

$$x\begin{pmatrix}2\\1\end{pmatrix} + y\begin{pmatrix}1\\3\end{pmatrix} = \begin{pmatrix}5\\5\end{pmatrix}$$

and $x = 2$, $y = 1$ does it: two steps of $a_1$, one of $a_2$.

## 2 / VECTORS AND WHAT WE CAN DO TO THEM

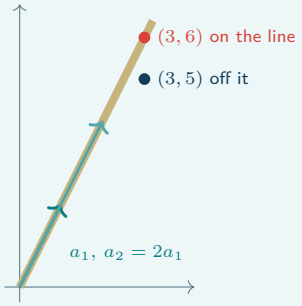
### Add vectors, scale them, and combine the two

| operation | rule | example with $u = (2, 1)$, $v = (1, 3)$ |
|---|---|---|
| sum | $u + v$ | $(2, 1) + (1, 3) = (3, 4)$ |
| scalar multiple | $3u$ | $3(2, 1) = (6, 3)$ |
| linear combination | $2u - v$ | $2(2, 1) - (1, 3) = (3, -1)$ |
| length | $\lvert u\rvert = \sqrt{u_1^2 + u_2^2}$ | $\lvert(2, 1)\rvert = \sqrt{5}$ |
| unit vector | $u/\lvert u\rvert$ | $(2, 1)/\sqrt{5}$ |

Adding vectors places them head to tail, and a linear combination $c_1u + c_2v$ is every point reachable by stretching each and adding.

## 3 / SPAN / WHICH RIGHT-HAND SIDES CAN BE REACHED

### Independent columns reach every point of the plane, and dependent columns only a line




For $A = \begin{pmatrix}1 & 2\\2 & 4\end{pmatrix}$ the second column is twice the first, so every combination $xa_1 + ya_2 = (x + 2y)(1, 2)$ lies on one line. Then $Ax = (3, 6)$ has infinitely many solutions, such as $x + 2y = 3$, while $Ax = (3, 5)$ has none. The columns of $\begin{pmatrix}2 & 1\\1 & 3\end{pmatrix}$ are independent, with determinant $5 \neq 0$, so every $b$ has exactly one solution.

## 4 / THE SAME IDEA IN THREE DIMENSIONS

### Three equations in three unknowns ask for $b$ as a combination of three columns in space

The span of one non-zero vector is a line, of two independent vectors a plane, of three independent vectors all of space.
Three equations therefore give one solution when the columns span space, none or infinitely many when they collapse to a plane or a line, and the vectors $\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$, $\mathbf{k} = (0, 0, 1)$ reach every point as $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

### ASIDE / ROWS OR COLUMNS

The row picture shows where the solution is, and the column picture shows what it is made of, which is the one that generalises.

### LINKS TO OTHER SHEETS

Elimination on these systems is in *Linear simultaneous equations and the LCM*; the next sheet is *Dot products, length and projection*.

**Convention.** Vectors are columns; $a_1, a_2$ are the columns of $A$; the span is the set of all linear combinations.